

Lecture 0: Introduction to Macro Public Finance

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Goal for this class

1. Develop skills and learn methods to write papers (theory and quantitative):
 - ▶ Learning by doing,
 - ▶ Going over proofs.
2. Create a repertoire of canonical papers and read thoroughly.
3. Getting you inspired and ready for your own research

Class logistics

- ▶ Meet once per week, up to 2H50mins sometimes. Short intermission.
- ▶ One in-class presentation.
- ▶ One referee report.
- ▶ Term paper/proposal at intersection of macroeconomics and public economics.
- ▶ Office Hours: NYU students and students writing a term paper, by private appointment.
- ▶ What I expect from you.

My research

I study the taxation of individuals and firms. I focus on three main issues:

1. The effect of taxes and government redistributive and insurance policies on lifecycle decisions: consumption, intensive labor supply, retirement.
 - ▶ How can we design the tax and Social Security system to incentive optimal retirement behavior?
2. The interplay between government tax incentives and private firms' compensation incentives: endogenous wages, performance pay.
 - ▶ How does redistribution affect wage contracts offered by firms?
3. The effects of taxes in imperfect markets with informational frictions and rents.

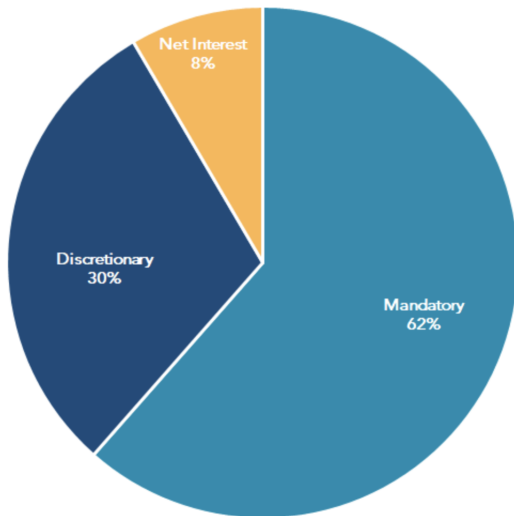
Why Macro?

- ▶ A few useful background facts:
- ▶ Government expenditures = $1/3$ GDP in the U.S.
 - ▶ Higher than 50% of GDP in some European countries
- ▶ Federal + State: Decentralization is a key feature of U.S. govt
 - ▶ One third of spending (10% of GDP) is done at state-local level (e.g. schools)
 - ▶ Two thirds (20% of GDP) is federal

Composition of Federal Government Spending: FY 2019

Fiscal year 2019

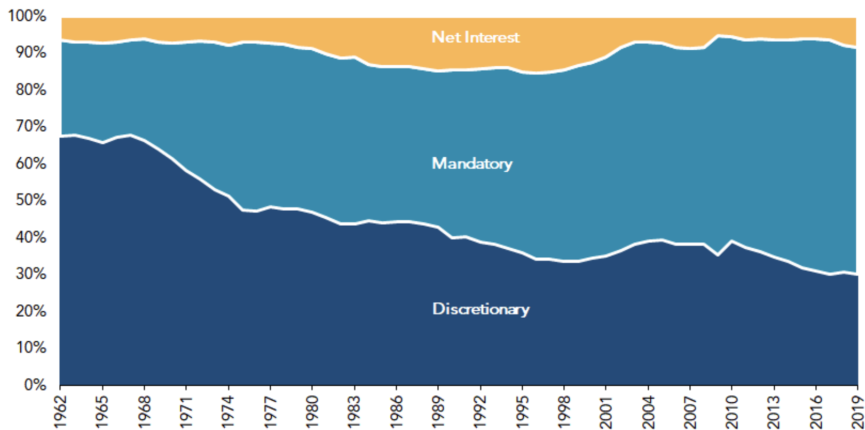
Share of total



Source: Congressional Budget Office. *The Budget and Economic Outlook: 2020 to 2030*, January 2020.

Composition of Federal Government Spending: FY 1962-2019

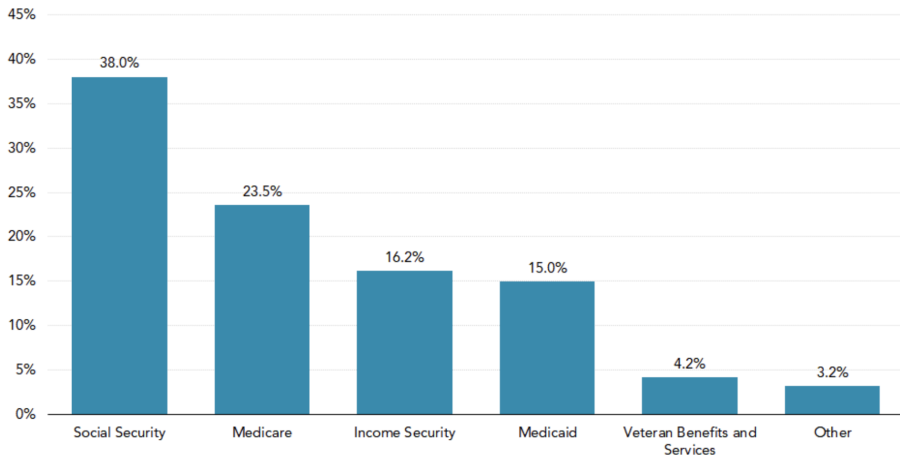
Share of total



Source: Congressional Budget Office. The Budget and Economic Outlook: 2020 to 2030, January 2020.

Composition of Federal Mandatory Spending: FY 2019

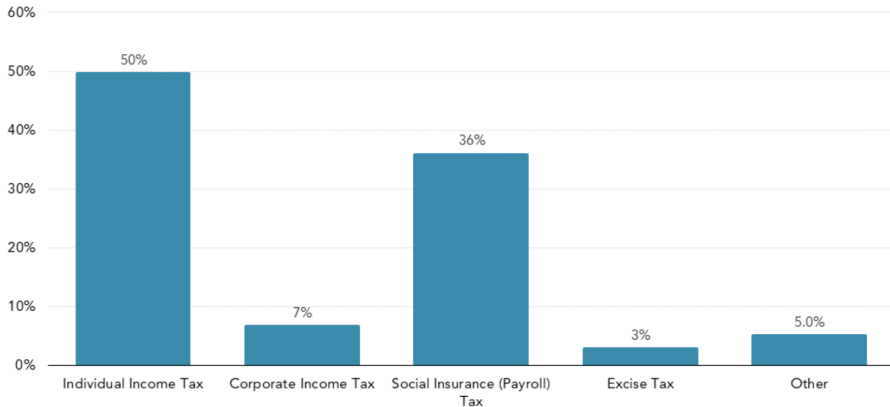
Share of total



Source: Office of Management and Budget. Historical Tables. Table 8.5, "Outlays for Mandatory and Related Programs: 1962–2025."

Sources of Federal Revenue: FY 2019

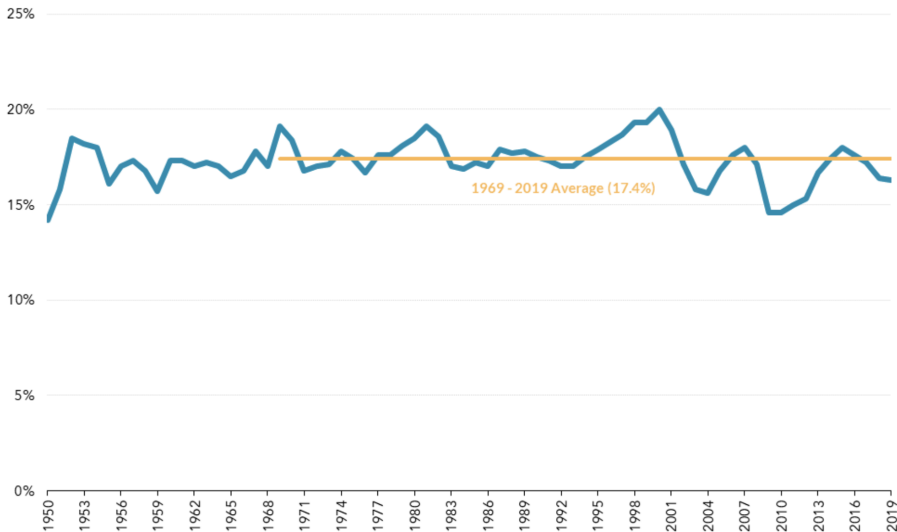
Share of total



Source: Office of Management and Budget. Historical Tables. Table 2.1, "Receipts by Source: 1934–2025."

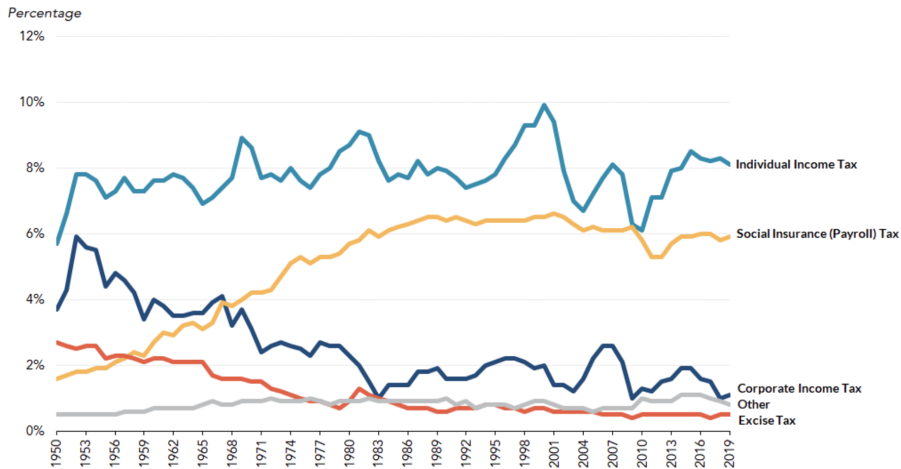
Federal Revenue as a Share of GDP: FY 1950-2019

Percentage



Source: Office of Management and Budget. Historical Tables. Table 2.3, "Receipts by Source as Percentages of GDP: 1934-2025."

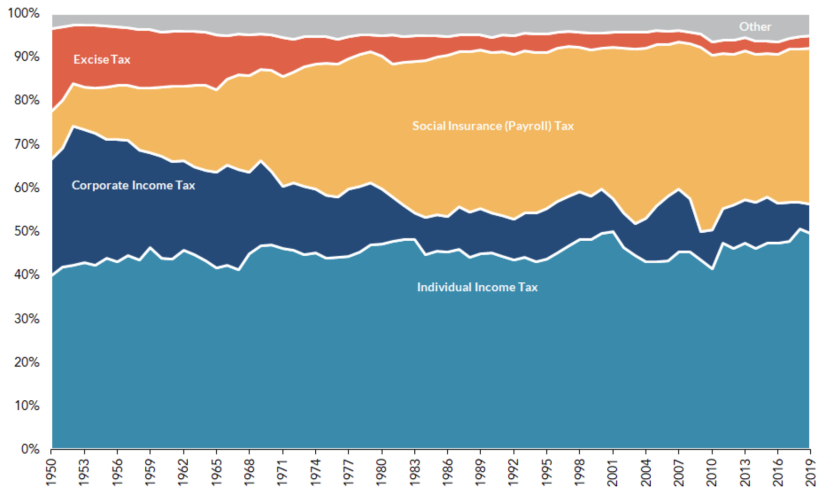
Sources of Federal Revenue as a Share of GDP: FY 1950-2019



Source: Office of Management and Budget, Historical Tables, Table 2.3, "Receipts by Source as Percentages of GDP: 1934-2025."

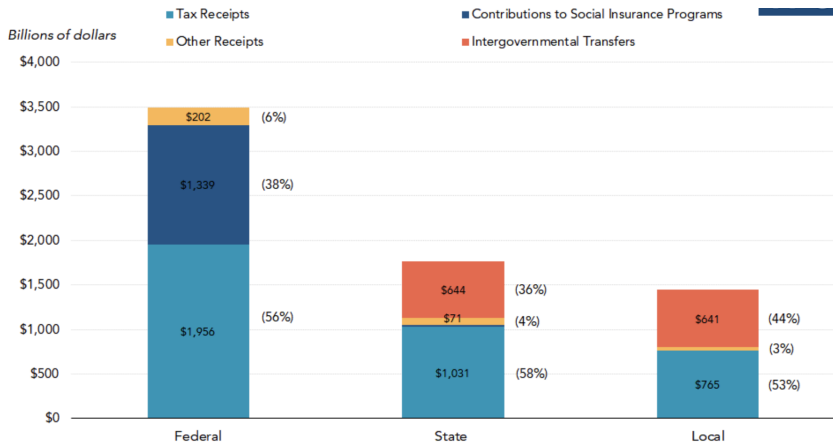
Sources of Federal Revenue: FY 1950-2019

Share of total



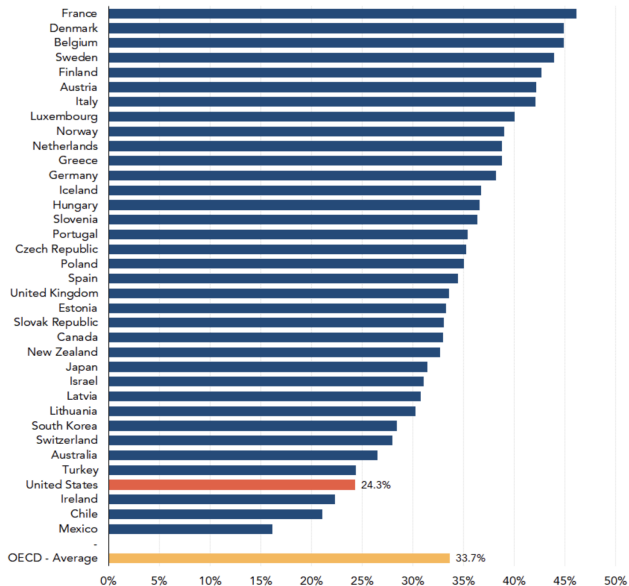
Source: Office of Management and Budget. Historical Tables. Table 2.1, "Receipts by Source: 1934-2025."

Sources of Federal, State, and Local Government Current Receipts: FY 2018



Source: Bureau of Economic Analysis. National Income and Product Accounts, 2020. Table 3.2, "Federal Government Current Receipts and Expenditures;" Table 3.20, "State Government Current Receipts and Expenditures;" Table 3.21, "Local Government Current Receipts and Expenditures."

Total Tax Revenue as a Share of GDP in OECD Countries: FY 2018



Source: OECD Revenue Statistics, Comparative Tables (retrieved 12.05.2019).

Notes: (a) Includes tax revenues from both federal and sub-national governments; b) The "OECD - Average" is weighted by GDP for all countries excluding the United States; (c) Data for Australia and Japan are for 2017.

Macro and Public Finance

- ▶ Public economics = Study of the role of the government in the economy
- ▶ Government is instrumental in most aspects of economic life:
 1. Government in charge of huge regulatory structure
 2. Taxes: governments in advanced economies collect 30-50% of National Income in taxes
 3. Expenditures: tax revenue funds traditional **public goods** (infrastructure, public order and safety, defense), and **welfare state** (education, retirement benefits, health care, income support)
 4. Taxes: governments in advanced economies collect 30-50% of National Income in taxes
 5. Macro-economic stabilization through central bank (interest rate, inflation control), fiscal stimulus, bailout policies

Bigger view on government

- ▶ Economists have a narrow minded view of individual behavior: selfish, rational, and utility based on own consumption only
- ▶ But social interactions are critical for humans: we naturally cooperate at many levels: families, communities, nation, states, global treaties
- ▶ Governments are a formal way to organize cooperation
- ▶ Archaic human societies depended on social cooperation for protection and taking care of the young, sick, and old
- ▶ Explains best why our modern nation states have defense, public health policies and provide education, health care, and retirement benefits
- ▶ Replacing social institutions by markets does not always work
- ▶ E.g., Retirement benefits: Saving for your own retirement is economically rational but in practice most people unable to do so unless institutions (employers/government) help them.

For Economists: Two General Rules for Government Intervention

1. **Failure of 1st Welfare Theorem:** Government intervention can help if there are market or individual failures. Markets first, government second. Why?
2. **Fallacy of the 2nd Welfare Theorem:** Distortionary government intervention is required to reduce economic inequality

Government Role 1: 1st Welfare Theorem Failure

- ▶ **1st Welfare Theorem:** If (1) no externalities, (2) perfect competition, (3) perfect information, (4) agents are rational, then private market equilibrium is Pareto efficient

- ▶ Government intervention may be desirable if:
 1. Externalities require government interventions (Pigouvian taxes/subsidies, public good provision)
 2. Imperfect competition requires regulation (typically studied in Industrial Organization)
 3. Imperfect or Asymmetric Information (e.g., adverse selection may call for mandatory insurance)
 4. Agents are not rational (= individual failures analyzed in behavioral economics, field in huge expansion): e.g., myopic or hyperbolic agents may not save enough for retirement

Government Role 2: 2nd Welfare Theorem Fallacy

- ▶ Even with no market failures, free market might generate substantial inequality. Inequality is an issue because people care about their relative situation.
 - ▶ US top income shares dropped dramatically from 1929 to 1950 and increased dramatically since 1980. Bottom 50% incomes have stagnated in real terms since 1980 [Piketty-Saez-Zucman '18]
- ▶ **2nd Welfare Theorem:** Any Pareto Efficient outcome can be reached by (1) Suitable redistribution of initial endowments [individualized lump-sum taxes based on indiv. characteristics and not behavior], (2) Then letting markets work freely
 - ▶ $\implies$ No conflict between efficiency and equity [**1st-Best Taxation**]
- ▶ Redistribution of initial endowments is not feasible (information pb)
 - ▶ $\implies$ govt needs to use **distortionary** taxes and transfers
 - ▶ $\implies$ Trade-off between efficiency and equity [**2nd-Best Taxation**]
- ▶ This class will focus on both roles.

Bibliography

Congressional Budget Office (2020). *The Budget and Economic Outlook: 2020 to 2030*, January 2020.

Office of Management and Budget. *Historical Tables*. Table 2.1, "Receipts by Source: 1934–2025."

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OECD Revenue Statistics, Comparative Tables (retrieved 12.05.2019).

Piketty, T., Saez, E., and Zucman, G. (2018). "Distributional National Accounts: Methods and Estimates for the United States." *Quarterly Journal of Economics*, 133(2), 553-609.